

AMENDMENT NO. 3 MARCH 1992
TO
IS 12015 : 1987 SPECIFICATION FOR CYPERMETHRIN,
TECHNICAL

(*Pages 6 and 7, clauses A-1.4 to A-1.6*) — Substitute the following for the existing clauses:

'A-1.4 Preparation of Internal Standard, Reference Standard and Sample Solutions

A-1.4.1 Weigh accurately 1 g of internal standard in a 100-ml volumetric flask and make up the volume using carbon tetrachloride-chloroform mixture [solvent system — 80 : 20 (v/v)]. This will give a 10 mg/ml solution of internal standard.

A-1.4.2 Weigh accurately 0.5 g of cypermethrin standard in a 100-ml volumetric flask and make up the volume using solvent system. This will give 5 mg/ml of standard solution. Take out 10.0 ml of this solution into a 50-ml volumetric flask with the help of a pipette. Add 5 ml of internal standard solution and mix well. Make up the volume to the mark with solvent system. Call this as Solution A.

A-1.4.3 Weigh accurately 0.5 g of sample in a 100-ml volumetric flask and make up the volume with solvent system. Take out 10.0 ml of this solution with a pipette into a 50-ml volumetric flask. Add 5 ml of internal standard solution, mix well. Make up the volume to the mark with solvent system. Call this as Solution B.

A-1.5 Procedure

A-1.5.1 Introduce 10 µl of Solutions A and B simultaneously into the HPLC unit. From the integrator, print out and note down peak areas of cypermethrin and internal standard in both cases. Calculate the cypermethrin content.

A-1.6 Calculation

Cypermethrin content,

$$\text{percent by mass} = \frac{m_1 \times A_2 \times A_3}{m_2 \times A_1 \times A_4} \times P$$

where

m_1 = mass, in g, of cypermethrin standard in Solution A;
 A_2 = peak area of cypermethrin in Solution B;
 A_3 = peak area of internal standard in Solution A;
 m_2 = mass, in g, of cypermethrin in sample in Solution;
 A_1 = peak area of cypermethrin standard in Solution A;
 A_4 = peak area of internal standard in Solution B ; and
 P = percent purity of cypermethrin standard.'

(Pages 7, 8 and 9, clauses A-2.2.1, A-2.4 to A-2.6) — Substitute the following for the existing clauses:

'A-2.2.1 *Gas Liquid Chromatograph (GLC)* — Equipped with flame ionization detector (FID) coupled to a printer-plotter-cum-integrator. The suggestive operative parameters are given below but can be changed in any other equipment, provided standardization is done:

Column	90 cm × 3 mm id Stainless steel packed with 3 percent Dexil 300 or suitable equivalent on chromosorb. W-HP, 100-120 mesh.
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Gas flow	
Carrier gas (N ₂)	40 ml/min
Fuel gas (H ₂)	40 ml/min
Air	300 ml/min

Temperature	
Column oven	240°C
Detector	270°C
Injection port	270°C

A-2.4 Preparation of Internal Standard, Reference Standard and Sample Solutions

A-2.4.1 Weigh accurately 2.0 g of internal standard in a 100-ml volumetric flask and make up the volume using toluene. This will contain 20 mg/ml of internal standard.

A-2.4.2 Weigh accurately 375 mg of cypermethrin standard in a 25-ml volumetric flask and make up the volume with toluene. Take out 10.0 ml of this solution with a pipette into another 25-ml volumetric flask, add 5.0 ml internal standard solution, make up the volume with toluene and mix well. Call this as Solution A. The solution will contain 6 mg/ml of cypermethrin.

A-2.4.3 Weigh accurately a quantity of sample containing equivalent to 375 mg of cypermethrin in a 25-ml volumetric flask and make up the volume with toluene. Take out 10.0 ml of this solution with a pipette into another 25-ml volumetric flask, add 5.0 ml of internal standard solution, make up the volume with toluene and mix well. Call this as Solution B. This solution will contain 6 mg/ml of cypermethrin.

A-2.5 Procedure

A-2.5.1 Inject 1.0 μ l of Solutions A and B simultaneously into the GLC unit and from the integrator, print out and note down peak areas of the internal standard, cypermethrin standard and the sample.

A-2.6 Calculation

Cypermethrin content,

$$\text{percent by mass} = \frac{M_1 \times A_2 \times A_3}{M_2 \times A_1 \times A_4} \times P$$

where

M_1 = mass, in g, of cypermethrin standard in Solution A;

A_2 = peak area of cypermethrin in Solution B;

A_3 = peak area of internal standard in Solution A;

M_2 = mass, in g, of cypermethrin in sample in Solution B;

A_1 = peak area of cypermethrin standard in Solution A;

A_4 = peak area of internal standard in Solution B; and

P = percent purity of cypermethrin standard.'

(Pages 9 and 10, clauses A-3.4.1 to A-3.5) — Substitute the following for the existing clauses:

'A-3.4.1 Standard Preparation — Weigh accurately 100 mg of standard cypermethrin in a 100-ml volumetric flask. Dissolve in chloroform and make up the volume to the mark. This solution will contain 1 000 μ g/ml of cypermethrin.

Take 5.0 ml of this solution into a 50-ml volumetric flask and dilute it to the mark with chloroform. The final solution will contain 100 μ g/ml of cypermethrin.

A-3.4.2 Solution Preparation — Weigh out accurately 100 mg of the sample in clean and dry 100-ml volumetric flask and make up the volume with chloroform. Take 5.0 ml of this solution in another 50-ml volumetric flask and make up the volume with chloroform. The final solution will contain 100 µg/ml of cypermethrin.

A-3.4.3 Determination — Fill the matched pair of cells with chloroform and measure cell error. It shall be less than 0.005 at 279 nm. Obtain the absorbance values at 279 nm of the standard and sample solutions in the cell against solvent blank. Correct the absorbance values for the cell error, if any, and calculate the cypermethrin content of the sample.

A-3.5 Calculation

Cypermethrin content,
percent by mass = $\frac{M_1 \times A_2}{M_2 \times A_1} \times P$
where

M_1 = mass, in g, of the cypermethrin standard;

A_2 = absorbance of the sample;

P = percent purity of the cypermethrin standard;

M_2 = mass, in g, of the sample; and

A_1 = absorbance of the standard.'

(FAD 1)